Cover page

University of British Columbia, Vancouver
Department of Computer Science

CPSC 304 Project Cover Page

Milestone #: 4_____

Date: 2025-08-02_____

Group Number: 33_____

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Project description

Our project stores performance statistics for League of Legends champions, including win rate, pick rate and ban rate. It helps players make informed in-game decisions by offering insights into champion performance.

Schema differences

Our schema is consistent with what we submitted in Milestones 2, other than some datatype changes.

Schema copy & screenshots of data

Amateur_player(UID: char, Rank: char, Preferred_role: char)

USERID	RANK	PREFERRED_ROLE
101	Bronze	Mid
102	Silver	Top
103	Gold	Support
104	Platinum	Jungle
105	Diamond	ADC

Esports_player1(UID, Rank, Team)

USERID	RANK	TEAM
201	Challenger	GenG
202	Challenger	T1
203	Challenger	HLE
204	Challenger	AL
205	Challenger	BLG

Esports_player2(Team, Position)

TEAM	POSITION
GenG	Top
T1	Jungle
HLE	Mid
AL	ADC
BLG	Support

Esports_player3(Team, StageName)

TEAM	STAGENAME
GenG	Kiin
T1	Oner
HLE	Zeka
AL	Hope
BLG	ON

Champions(ChampionName: char, Tier: char, Archetype: char, **RoleName**: char)

CHAMPIONNAME	ARCHETYPE	TIER	ROLENAME
Ambessa	Fighter	OP	Top
Pantheon	Fighter	1	Jungle
Azir	Mage	1	Mid
Ezreal	Marksman	2	ADC
Alistar	Tank	OP	Support
Kai'Sa	Marksman	1	ADC

Stats(Winrate: decimal, Pickrate: decimal, **ChampionName**: char, Banrate: decimal)
Role(RoleName: char, Map_area: char)

CHAMPIONNAME	WINRATE	PICKRATE	BANRATE
Alistar	54.69	19.55	4.2
Ezreal	47.14	11.72	2.5
Azir	52.23	10.44	2.38
Pantheon	52.04	16.51	19.58
Ambessa	50.94	13.67	5.88

Matchup(**ChampionName**: char, Counter-Champion: char, **PatchID**: DECIMAL(4,2), MatchupWinrate: decimal, Matches Played: integer)

CHAMPIONNAME	COUNTER_CHAMPION	PATCHID	MATCHUPWINRATE	MATCHESPLAYED
Ambessa	Gwen	25.14	46.15	39
Pantheon	Trundle	25.14	51.35	37
Azir	Taliyah	25.14	42.55	47
Kai'Sa	Lucian	25.13	43.33	30
Alistar	Braum	25.13	39.39	33

Build(SkillOrder: char, **ChampionName**: char, **PatchID**: char)

CHAMPIONNAME	SKILLORDER	PATCHID
Ambessa	QEW	25.14
Alistar	QWE	25.13
Ezreal	QEW	25.12
Azir	WQE	25.11
Pantheon	QWE	25.1

Item(ItemName: char, Effect: char)

NAME	EFFECT
Infinity Edge	Massively enhances critical strikes.
Luden's Companion	Grants Ability Power, Mana, and burst damage.
Heartsteel	Grants stacking Health and HP scaling damage.
Redemption	ACTIVE: Heals allies and damages enemies in a circle.
Zhonya's Hourglass	ACTIVE: Become invincible for the duration but unable to take action.
Eclipse	Grants Attack damage and a shield.

Build-Uses-Item(**SkillOrder**: char, **ChampionName**: char, **ItemName**: char)
Screenshot

CHAMPIONNAME	SKILLORDER	ITEMNAME
Alistar	QWE	Redemption
Ambessa	QEW	Heartsteel
Azir	WQE	Zhonya's Hourglass
Ezreal	QEW	Eclipse
Pantheon	QWE	Infinity Edge

Patch(PatchID: char, date: char)

PATCHID	PATCHDATE
25.14	2025 - 07 - 15
25.13	2025 - 06 - 24
25.12	2025 - 06 - 10
25.11	2025 - 05 - 27
25.1	2025 - 05 - 13

Screenshots demonstrating the functionality of each query using the GUI (INSERT and DELETE)

Query: INSERT:

Insert Values

Inserts a new Matchup.

ChampName: Azir

counterChamp: Ambessa

patchID: 25.10

winrate: 20.00

matches: 200

Insert

CHAMPIONNAME	COUNTER_CHAMPION	PATCHID	MATCHUPWINRATE	MATCHESPLAYED
Ambessa	Gwen	25.14	46.15	39
Pantheon	Trundle	25.14	51.35	37
Azir	Taliyah	25.14	42.55	47
Kai'Sa	Lucian	25.13	43.33	30
Alistar	Braum	25.13	39.39	33

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Azir	Taliyah	25.14	42.55	47
Kai'Sa	Lucian	25.13	43.33	30
Alistar	Braum	25.13	39.39	33
Azir	Ambessa	25.1	20	200

Query: DELETE:

Delete Champion by their ChampionName in Champions

Deletes a role. Will cascade and delete all champions of that role.

Role to Delete (Top, Mid...): Top

Top

Delete

Display Table Contents

Select a table from the dropdown to get its contents/p>

Select a table: Amateur_player Display Table

Role DELETED successfully!

CHAMPIONNAME	ARCHETYPE	TIER	ROLENAME
Ambessa	Fighter	OP	Top
Pantheon	Fighter	1	Jungle
Azir	Mage	1	Mid
Ezreal	Marksman	2	ADC
Alistar	Tank	OP	Support
Kai'Sa	Marksman	1	ADC

Display Table Contents

Select a table from the dropdown to get its contents/p>

Select a table: Amateur_player Display Table

CHAMPIONNAME	ARCHETYPE	TIER	ROLENAME
Pantheon	Fighter	1	Jungle
Azir	Mage	1	Mid
Ezreal	Marksman	2	ADC
Alistar	Tank	OP	Support
Kai'Sa	Marksman	1	ADC

List of all SQL queries used:

Project Operation:project the Matchup table to find counter-champion and relevant winrate.

(file: projection.sql)

Join Operation: join the Champions table and the Stats table to find winrate, pickrate and banrate of all champions.(file: join.sql)

Aggregation with Group By: group the Champions table to count number of champions in each tier.(file: AggregationWithGroupBy.sql)

Aggregation with Having: find champions who have more than one build.(file: AggregationWithHaving.sql)

Nested Aggregation with Group By: find champions whose average matchup winrate is higher than the average of all champion's matchup winrates.(file: Nested AggregationWithGroupBy.sql)

Division: find champions that have a counter in every patch.(file: Division.sql)